

parking spaces available and not the actual feasible parking space. Hence, this automated system was designed that could self-park based on the available spaces. Coordinates systems was used to detect available car parking spaces to indicate the regions of interest and a car classifier. This paper shows that the initial work done here has an accuracy that ranges from 90% to 100% for a 4 space car park. The work done showed that the utilization of a dream based vehicle leave the board framework would most likely recognize and demonstrate the accessible vehicle park spaces. Sensors alongside a camera were introduced to give accessible parking spots. In view of the info video, Haar-like highlights decided the nearness of a vehicle inside the parking spot. The object classification was done using a tool called OpenCV. Two sets of input images are necessary to train this algorithm. One set called the positive images contain the object to be detected and another set called the negative images do not contain the object to be detected. Once this data was analyzed, coordinate system was used to detect the coordinate of the car parking spaces with the help of a marker tool of OpenCV. This was used to determine the number of parking spaces available and the best place for the car to park in the list of available spaces.

This system was designed[3] using advanced technologies where the data streams included IMU, GPS, CAN messages and HD video streams of the driver's face, driver's cabin, the forward path and the instrument cluster. It implements "naturalistic driving study", which is not restricted by experimental design and which collects audio, video and other elements of driving over a large period of time and to use this collected data to help the autonomous car. The aim of this paper is to understand the data

being collected throughout from vehicles in order to design, implement and run the autonomous system, inform the providers of insurance of safety measures, and enlighten the educational institutions on how to use these autonomous cars in order to reduce the burden of mankind. It aims at understanding human behaviour in a semi-automated system for a period of 1 year and using those reactions to drive the autonomous car. It uses AI to inspect the driving experience through the collection of data and to use human expertise to understand how to react in difficult situations.

In this study [4], it is stated that the project mainly focuses on the input which will be the video/image data which is continuously captured with the help of a webcam interfaced to the Raspberry Pi.

It detects the object and tracks that object by moving the camera in the direction of the detected object. The Raspberry pi offers better size but less speed. Accuracy of both systems is similar even if the FPS rate is very different. In this framework we can utilize the foundation subtraction by utilizing the fixed camera by producing the frontal area cover. It contrasts the casing and typical one with foundation pictures or model which has contain the static piece of the scene, everything is considered as the foundation part of pictures as a rule. In this background subtraction can done with the raspberry pi camera. In this system, the image taken by the camera was sent to the raspberry pi, to which the camera was connected using an USB port. The input image was executed on the Raspbian OS software followed by the execution of the python code. The signals generated after the code is executed, is sent to the car. The car then detects objects and the path to the destination effectively and starts driving towards it. Its components include